

Programming Assignment-1

Monsoon 2026

Kindly note the following instructions before starting the assignment.

- This is a group assignment.
- You have to attempt any one of two questions.
- Kindly note that the following problems can be solved using brute force. If a team uses brute-force, they should program in C (any other language will lead to negative marks). C, C++, Python programming languages are allowed if the submission uses the Lemke–Howson / any other algorithm.

1 PSNE and Very Weakly Dominant Strategies for n -Player Game

Given an n -player game, list all PSNE and very weakly dominant strategies for each player. We are given n -the number of players, $|S_i|$ -the number of strategies for each player $i \in \{1, 2, \dots, n\}$.

1.1 Input Format

- First line contains the number of players n .
- Second line contains n numbers, i^{th} number representing $|S_i|$.
- Next line contains the payoffs in the NFG format (described in the Gambit Project).

1.2 Output Format

- First line should contain the number of PSNE (denoted by n_{psne}).
- Followed by n_{psne} lines, the i^{th} line containing n space-separated numbers corresponding to the equilibrium strategies for each player respectively.
- Next, you should output n lines, with the i^{th} line listing the number of very weakly dominant strategies for the i^{th} player followed by the dominant strategies.

1.3 Constraints

- $n \prod_i |S_i| \leq 10^6$.
- 60 seconds per test case.
- Memory limit: 1GB.

2 Find All Dominated Strategies of a Bimatrix Game

Given a bimatrix game, find all dominated strategies (strongly as well as weakly) for both players.

2.1 Input

The input is a bimatrix game with the payoffs listed in the NFG format (as described in the Gambit Project).

- First line contains the number of strategies for the row player (R).
- Second line contains the number of strategies for the column player (C).
- The third line contains the list of payoffs in the NFG format.

2.2 Output

- The first line must contain the number of strongly dominated strategies for the row player, followed by the space-separated indices of these strategies.
- The second line must contain the number of weakly dominated strategies for the row player, followed by the space-separated indices of these strategies.
- The third line must contain the number of strongly dominated strategies for the column player, followed by the space-separated indices of these strategies.
- The fourth line must contain the number of weakly dominated strategies for the column player, followed by the space-separated indices of these strategies.

2.3 Constraints

- $R \times C \leq 25$.
- Time limit: 60 seconds.
- Memory limit: 1GB.